

Math 118 – Fourier Analysis

Homework 4 — my solutions

Fall 2026 ■ Keshav Ramamurthy

HOMEWORK 4
due Sat, October 3, 11:59 PM
(Upload your solutions to Gradescope)

1. (5 points) (#1 p. 83.) Expand the function $f(x) = x^2$ in a Fourier series valid on the interval $-\pi \leq x \leq \pi$. Use a computer to plot both f and the partial sums of its Fourier series,

$$S_N(x) = a_0 + \sum_{k=1}^N \left(a_k \cos(kx) + b_k \sin(kx) \right)$$

for $N = 1, 2, 5, 7$. Observe how the graphs of the partial sums $S_N(x)$ approximate the graph of f . Plot the same graphs over the interval $-2\pi \leq x \leq 2\pi$.

2. (4 points) (#8 p. 84.) Expand the function

$$f(x) = \begin{cases} 1 & -1/2 < x \leq 1/2, \\ 0 & -1 < x \leq -1/2 \text{ or } 1/2 < x \leq 1, \end{cases}$$

in a Fourier series valid on the interval $-1 \leq x \leq 1$. Use a computer to plot the various partial Fourier series along with the graph of f as in exercise 1 for $N = 5, 10, 20$ and 40 terms. Notice how much slower the series converges to f in this example than in exercise 1. What accounts for the slow rate of convergence in this example?

3. (6 points) (#20 p. 85.) If $f(x)$ is continuous on the interval $0 \leq x \leq a$, show that its even periodic extension is continuous everywhere. Does this statement hold for the odd periodic extension? What additional condition(s) is (are) necessary to ensure that the odd periodic extension is everywhere continuous?
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4. (2 points) (#19 p. 85.) Let $F(x)$ be the (pointwise sum of the infinite) Fourier series for the function

$$f(x) = \begin{cases} 1 & -1/2 < x \leq 1/2, \\ 0 & -1 < x \leq -1/2 \text{ or } 1/2 < x \leq 1. \end{cases}$$

State the precise value of $F(x)$ for each x in the interval $-1 \leq x \leq 1$.

5. (3 points) (#22 p. 85.) In exercise 1, you found the Fourier series for the function $f(x) = x^2$, $-\pi \leq x \leq \pi$. Explain why this series converges uniformly to x^2 on $[-\pi, \pi]$. Use this to show that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. (*Hint:* What happens at $x = \pi$?)

Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggess and Narcowich.

Solutions

Problem 1

Solution. We start with $f(x) = x^2$. Since it is even, $b_k = 0$. The constant term is

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{\pi^2}{3}.$$

For $k \geq 1$, since it's even we have

$$a_k = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(kx) dx.$$

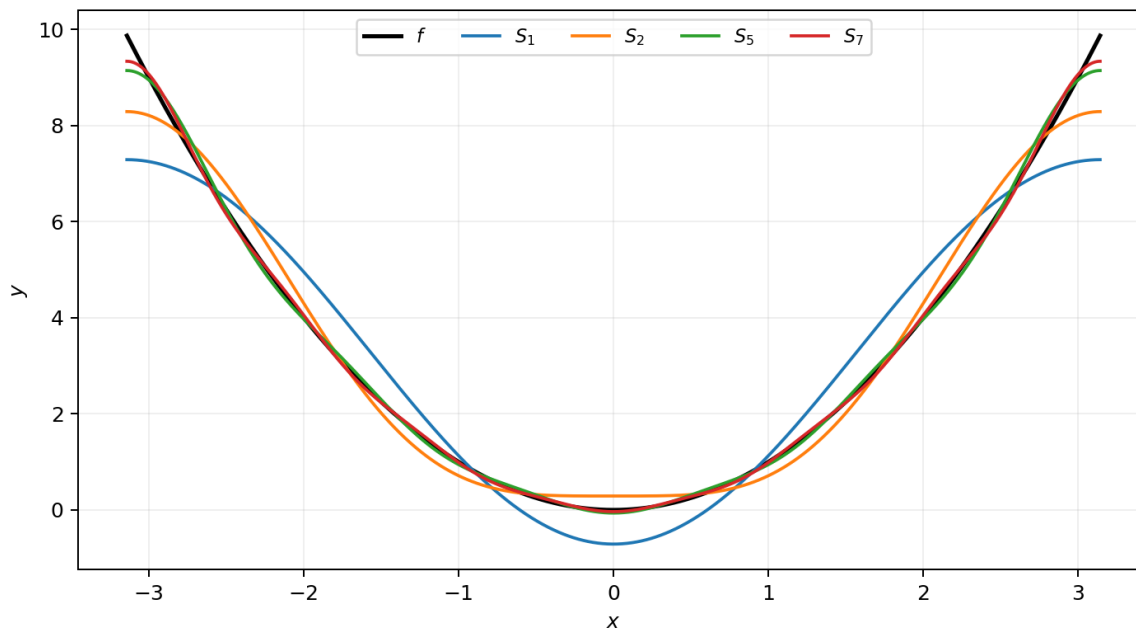
Integrating by parts,

$$\int_0^{\pi} x^2 \cos(kx) dx = -\frac{2}{k} \int_0^{\pi} x \sin(kx) dx = \frac{2\pi(-1)^k}{k^2},$$

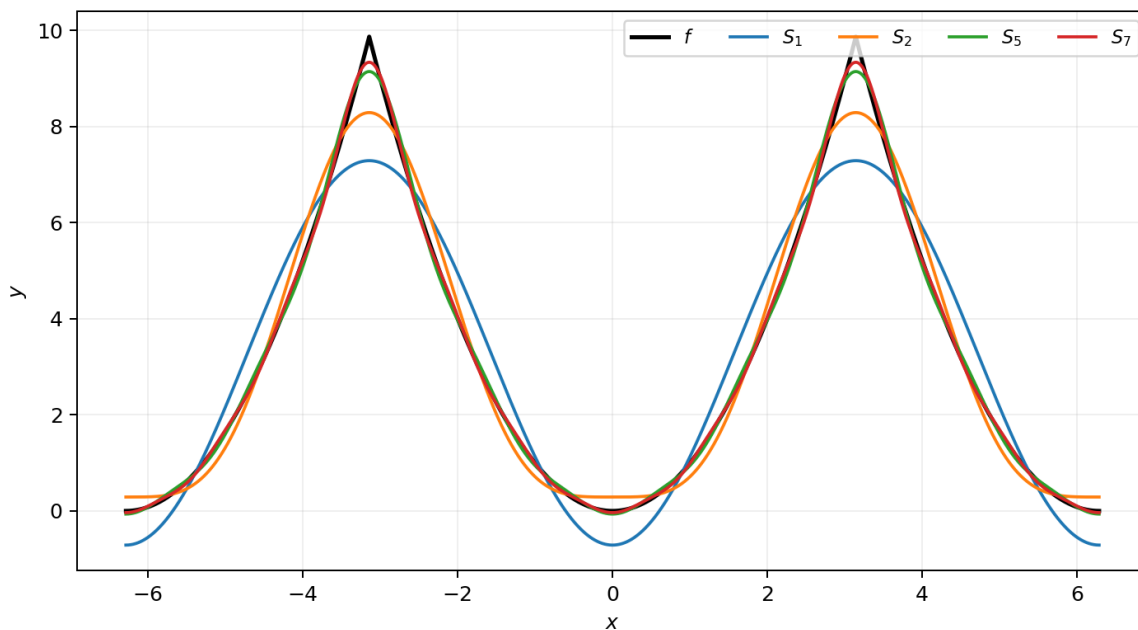
so we have that $a_k = 4(-1)^k/k^2$. Thus

$$S_N(x) = \frac{\pi^2}{3} + 4 \sum_{k=1}^N \frac{(-1)^k}{k^2} \cos(kx).$$

For $N = 1, 2, 5, 7$, the partial sums approach x^2 on $[-\pi, \pi]$.



Over $[-2\pi, 2\pi]$, the series repeats the graph from $[-\pi, \pi]$.



□

Problem 2

Solution. On $[-1, 1]$ the frequencies are $k\pi$, and Since f is even, $b_k = 0$, so the constant term is

$$a_0 = \frac{1}{2} \int_{-1}^1 f(x) dx = \frac{1}{2} \int_{-1/2}^{1/2} 1 dx = \frac{1}{2}.$$

For $k \geq 1$ we have that

$$a_k = \int_{-1}^1 f(x) \cos(k\pi x) dx = \int_{-1/2}^{1/2} \cos(k\pi x) dx = \frac{2 \sin(k\pi/2)}{k\pi}.$$

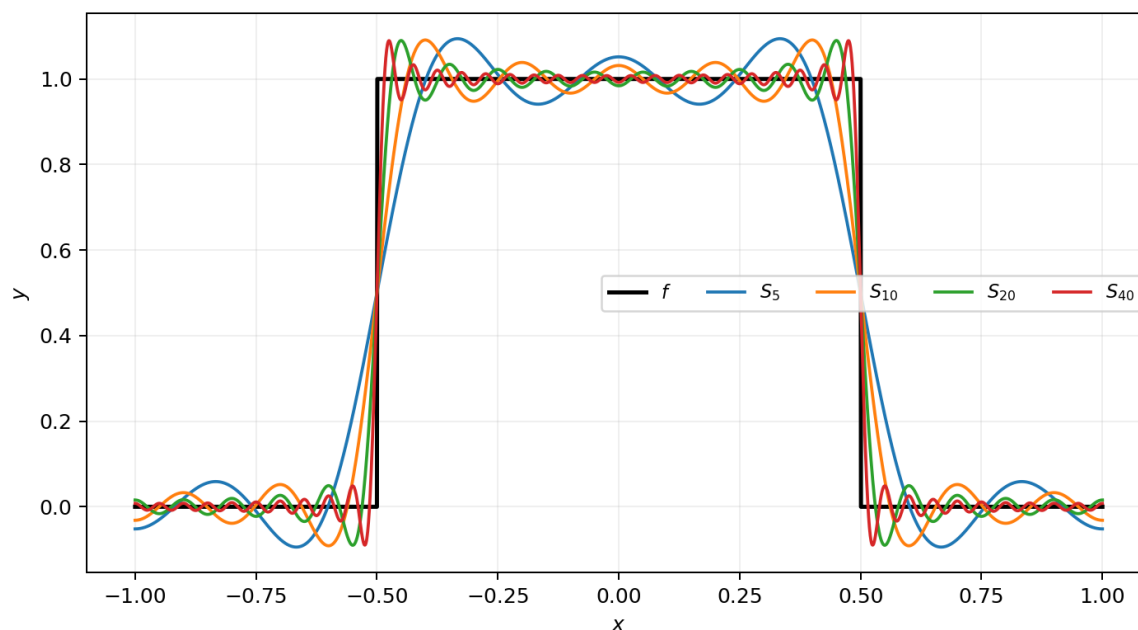
Plugging these back in, we have

$$F(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\sin(k\pi/2)}{k} \cos(k\pi x), \quad S_N(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^N \frac{\sin(k\pi/2)}{k} \cos(k\pi x).$$

Only odd values of k appear, since $\sin(k\pi/2) = 0$ for even k .

The graph jumps from 0 to 1 at $-1/2$ and back to 0 at $1/2$.

The partial sums overshoot near those points, and the nonzero coefficients have factor $1/k$, rather than $1/k^2$ as in Problem 1, so this series converges more slowly.



□

Problem 3

Solution. For the even extension, we reflect f across 0 and repeat every $2a$.

The only possible breaks are at 0 or the endpoints where two meet. At 0, both sides go to $f(0)$ and at $-a$ and a , they both go to $f(a)$.

For any other point where two points meet it is analogous to the endpoints at $-a$ and a and thus the even extension is continuous.

For the odd extension, the reflected half is $-f(-x)$. The two limits at 0 are $-f(0)$ and $f(0)$, and the limits at the ends of a period are $-f(a)$ and $f(a)$.

Each pair only has equal values exactly when $f(0) = f(a) = 0$, and thus continuity only occurs when $f(0) = f(a) = 0$. □

Problem 4

Solution. The only jumps are at $x = \pm 1/2$. Besides them, the fourier series converges to 1 in the middle interval and 0 on the outer intervals. At $x = \pm 1$, it also converges to 0, since the adjacent periods meet at 0.

At each jump, the left and right values are 0 and 1 in some order. Their average is $\frac{0+1}{2} = \frac{1}{2}$, so

$$F(x) = \begin{cases} 1, & -\frac{1}{2} < x < \frac{1}{2}, \\ \frac{1}{2}, & x = -\frac{1}{2} \text{ or } x = \frac{1}{2}, \\ 0, & -1 \leq x < -\frac{1}{2} \text{ or } \frac{1}{2} < x \leq 1. \end{cases}$$

□

Problem 5

Solution. From Problem 1, the partial sums are

$$S_N(x) = \frac{\pi^2}{3} + 4 \sum_{k=1}^N \frac{(-1)^k}{k^2} \cos(kx).$$

The values at $-\pi$ and π are both π^2 , so the repeated pieces meet at the same value. Each piece is a quadratic, and there are no jumps where they meet. The Fourier series therefore converges to x^2 at every point of $[-\pi, \pi]$, including the endpoints.

Since $|\cos(kx)| \leq 1$, the size of the tail after S_N is bounded for every x by

$$4 \sum_{k=N+1}^{\infty} \frac{1}{k^2}.$$

This tends to 0 as $N \rightarrow \infty$, so the convergence is uniform on $[-\pi, \pi]$.

Now set $x = \pi$. Since $\cos(k\pi) = (-1)^k$, each term in the sum is positive. Taking the limit gives

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{k=1}^{\infty} \frac{1}{k^2} \quad \rightarrow \quad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

□